Pandit Deendayal Energy University, Gandhinagar
School of Technology
Practical End Semester Examination – April-May 2025
B. Tech. – Computer Science and Engineering
Semester – VI

Subject: Big Data Analytics LAB **Date: 30/04/2025**
Course code: 23CP309P **Time: 11:00 am to 1:30 PM**

SET1

Practical Exam: Spark Data Processing Pipeline

Instructions: Complete all six parts of this question. Implement your solutions in Apache Spark using Scala. Each part is worth 5 marks.

You are provided with a dataset in the form of a text file containing customer transactions. Each line in the file represents a transaction with the following format:

customer_id, transaction_date, transaction_amount, product_category
123, 2024-10-01, 50.5, Electronics
124, 2024-10-02, 20.0, Books
123, 2024-10-03, 100.0, Electronics
125, 2024-10-01, 75.0, Clothing

1. **Data Loading and Basic Filtering**
   - **a.** Load the dataset as an RDD or DataFrame in Spark. **(2 Marks)**
   - **b.** Filter out any transactions with missing values or where transaction_amount is zero or negative. **(3 Marks)**
2. **Total Expenditure by Customer**
   Calculate the total transaction_amount spent by each customer across all transactions and output the results in the format (customer_id, total_expenditure). **(5 Marks)**
3. **Average Transaction Amount by Product Category**
   Compute the average transaction_amount for each product_category. The output should be in the format (product_category, average_transaction_amount). **(5 Marks)**
4. **Top 3 Customers by Expenditure**
   Identify the top 3 customers who have spent the most. Output the results in the format (customer_id, total_expenditure). **(5 Marks)**
5. **Transaction Count per Day**
   Calculate the total number of transactions for each date (transaction_date). The output should be in the format (transaction_date, transaction_count). **(5 Marks)**
6. **Most Popular Product Category**
   Determine the product category with the highest total number of transactions. Output the result in the format (product_category, transaction_count).**(5 Marks)**

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Set II

Practical Exam: Spark Processing on Product Reviews Dataset

Instructions: Complete all six parts of this question. Implement your solutions in Apache Spark using Scala. Each part is worth 5 marks.

You are provided with two datasets:

Reviews Dataset: Each line represents a product review with the format:
review_id, product_id, user_id, review_date, rating, review_text
R1, P123, U456, 2024-10-01, 4.5, "Great product, very useful!"
R2, P124, U457, 2024-10-02, 3.0, "Average quality, expected better."
R3, P123, U458, 2024-10-03, 5.0, "Excellent, highly recommended!"
Products Dataset: Each line contains information about a product with the format:
product_id, product_name, product_category, price
P123, "Wireless Mouse", "Electronics", 25.0
P124, "Hardcover Notebook", "Stationery", 10.0

1. **Data Loading and Basic Filtering**
   Load both datasets as RDDs or DataFrames. Filter out reviews with missing values or ratings less than 1.0. Only include reviews for products that exist in the Products dataset. **(5 Marks)**
2. **Average Rating by Product**
   Calculate the average rating for each product. Output the results in the format (product_id, average_rating). **(5 Marks)**
3. **Most Reviewed Products**
   Identify the top 5 products with the highest number of reviews. Output the results in the format (product_id, review_count). **(5 Marks)**
4. **Join Datasets to Include Product Information**
   Join the Reviews dataset with the Products dataset based on product_id. Display the combined data in the format (product_name, review_date, rating, user_id, review_text). **(5 Marks)**
5. **Average Rating by Category**
   Calculate the average rating for each product category (e.g., Electronics, Stationery). Output the results in the format (product_category, average_rating). **(5 Marks)**
6. **Sentiment Analysis on Review Text**
   For each product, calculate the percentage of reviews that contain positive keywords such as "great," "excellent," or "recommended." Use this as a basic sentiment score for each product. Format the output as (product_id, positive_review_percentage). **(5 Marks)**

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SET1

Provided to you a sample dataset to perform the operations mentioned below:

```
val data = Seq(
  (1, 101, 1001, 50, "2024-01-01", "New York"),
  (2, 102, 1002, 75, "2024-01-02", "Los Angeles"),
  (3, 103, 1001, 100, "2024-01-03", "Chicago"),
  (4, 104, 1003, 120, "2024-01-04", "Houston"),
  (5, 105, 1002, 80, "2024-01-05", "New York"),
  (6, 106, 1004, 90, "2024-01-06", "Chicago"),
  (7, 107, 1005, 110, "2024-01-07", "Los Angeles"),
  (8, 108, 1006, 130, "2024-01-08", "Houston"),
  (9, 109, 1004, 70, "2024-01-09", "New York"),
  (10, 110, 1007, 85, "2024-01-10", "Chicago"),
  (11, 111, 1005, 95, "2024-01-11", "Los Angeles"),
  (12, 112, 1008, 105, "2024-01-12", "Houston"),
  (13, 113, 1006, 115, "2024-01-13", "New York"),
  (14, 114, 1009, 125, "2024-01-14", "Chicago"),
  (15, 115, 1010, 135, "2024-01-15", "Los Angeles")
)
val schema = Seq("transaction_id", "product_id", "customer_id", "amount",
"transaction_date", "city")
```

Perform Data Manipulation and Analysis with PySpark SQL (10 marks)

a. Write a PySpark SQL query to calculate the total sales amount for each city. (3 marks)
b. Extend the previous query to find the top 3 cities where the highest sales were made. (3 marks)
c. Write a PySpark SQL query to calculate the average transaction amount per month for the year 2024. (4 marks)

Data Filtering and Aggregation with Scala Spark SQL (10 marks)

a. Write a Scala Spark SQL query to filter out transactions where the transaction date is after January 6, 2024. (3 marks)

b. Extend the previous query to calculate the total expenditure for each city. (4 marks)

c. Write a Scala Spark SQL query to find the average transaction amount for transactions made in Chicago. (3 marks)

Advanced Analytics with PySpark SQL (10 marks)

a. Write a PySpark SQL query to calculate the average total amount spent by customers in each city. (5 marks)

b. Write a PySpark SQL query to identify the city where the highest single transaction amount occurred. (5 marks)

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SET II

You are provided with a dataset representing e-commerce transactions. The dataset contains the following columns:

- **order_id**: Unique identifier for each order.
- **product_category**: Category of the product purchased.
- **customer_id**: Unique identifier of the customer who made the purchase.
- **amount**: Total amount spent on the order in dollars.
- **order_date**: Date the order was placed.
- **region**: The geographical region where the customer resides.

```
val data = Seq(
  (1, "Electronics", 101, 120, "2024-01-01", "North America"),
  (2, "Clothing", 102, 75, "2024-01-02", "Europe"),
  (3, "Home Appliances", 103, 250, "2024-01-03", "Asia"),
  (4, "Books", 104, 30, "2024-01-04", "Africa"),
  (5, "Clothing", 105, 60, "2024-01-05", "North America"),
  (6, "Electronics", 106, 100, "2024-01-06", "Europe"),
  (7, "Electronics", 107, 150, "2024-01-07", "Asia"),
  (8, "Furniture", 108, 500, "2024-01-08", "South America"),
  (9, "Books", 109, 40, "2024-01-09", "North America"),
  (10, "Furniture", 110, 700, "2024-01-10", "South America")
)
```

val schema = Seq("`order_id` ", " `product_category` ", " `customer_id` ", "amount", " `order_date` ", "region")

1. Data Manipulation and Analysis with PySpark SQL (10 marks)

a. Write a PySpark SQL query to calculate the total sales amount for each product category. **(3 marks)**
b. Extend the previous query to find the top 3 product categories where the highest sales were made. **(3 marks)**
c. Write a PySpark SQL query to calculate the average transaction amount per region for the year 2024. **(4 marks)**

2. Data Filtering and Aggregation with Scala Spark SQL (10 marks)

a. Write a Scala Spark SQL query to filter out transactions where the order date is after January 6, 2024. **(3 marks)**
b. Extend the previous query to calculate the total expenditure for each region. **(4 marks)**
c. Write a Scala Spark SQL query to find the average transaction amount for orders in the "Electronics" category. **(3 marks)**

3. Advanced Analytics with PySpark SQL (10 marks)

a. Write a PySpark SQL query to calculate the total sales amount for each customer and identify the top 3 highest-spending customers. **(5 marks)**
b. Write a PySpark SQL query to identify the region where the highest single order amount occurred. **(5 marks)**